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Economics 144
Economic Forecasting
Winter, 2020

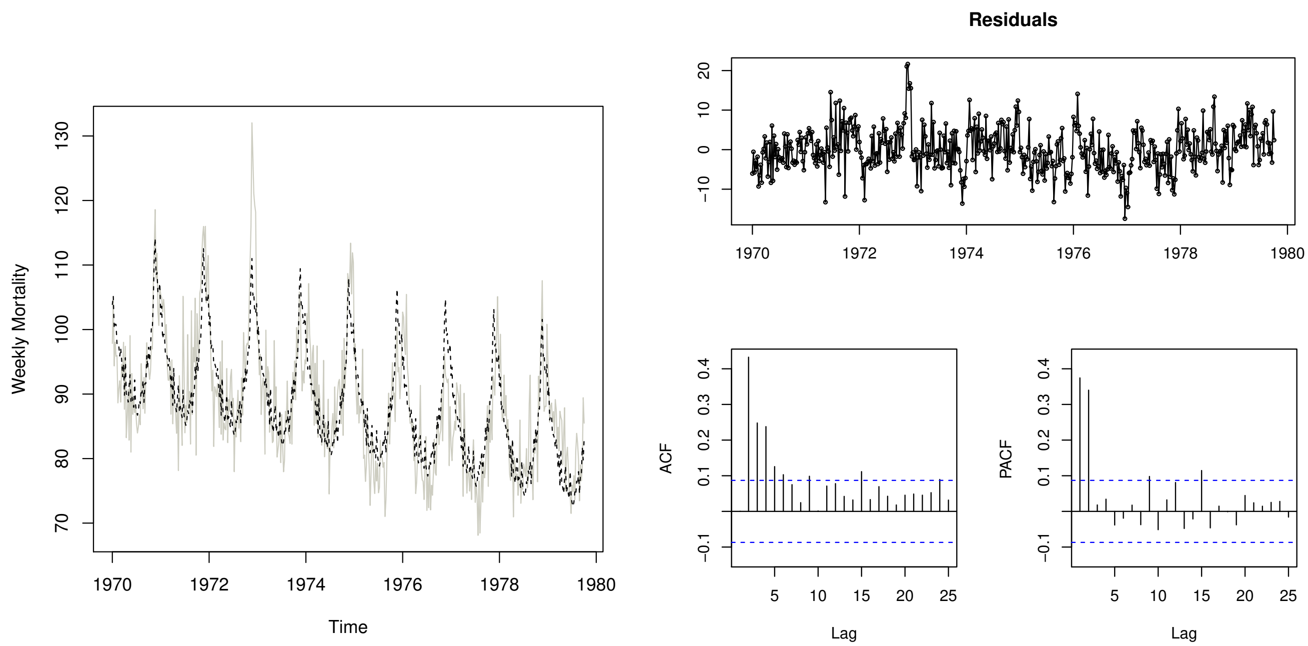
Midterm Exam
February 13, 2020

For full credit on a problem, you need to show all your work and the formula(s) used.

First Name	
Last Name	
UCLA ID #	

Please do not start the exam until instructed to do so.

1.(15%) For this problem the data consist of the average weekly cardiovascular mortality in Los Angeles County from 1970-1979 as shown in the left-figure below. We then fit a model with trend and seasonality components to the data (left-figure, the dashed line), and plot the respective residuals (right-figure), and the ACF and PACF of the residuals (two bottom-right plots).



(a) Based on the plots above, how would you improve the model fit?

(b) Does the original data series seem to be covariance stationary? If so, how can you tell? If not, why not and what would you recommend to make it covariance stationary.

(c) Does the residuals plot show any serial correlation? Explain your answer in detail.

(d) Suppose you use `auto.arima` to fit a model to the original data series, and you get the following output:

ARIMA(2,1,1)(2,0,2)[52]							
Coefficients:							
	ar1	ar2	ma1	sar1	sar2	sma1	sma2
	0.3285	0.2939	-0.9822	0.2424	0.4687	-0.2648	-0.2758
s.e.	0.0480	0.0422	0.0119	0.0632	0.0628	0.0753	0.0882

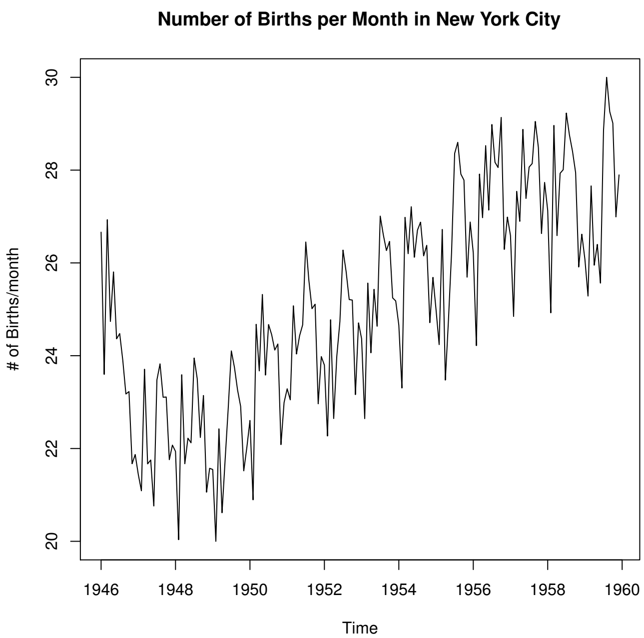
Provide a detailed interpretation of the ARIMA model suggested by R. Make sure to comment on any trend, seasonality, and/or cycles if appropriate.

2.(15%) (a) Describe a difference between a forecast for an MA process vs. a forecast for an AR process.

(b) Describe the advantage of combining forecasts (over e.g., a single forecast) and when would you consider combining them.

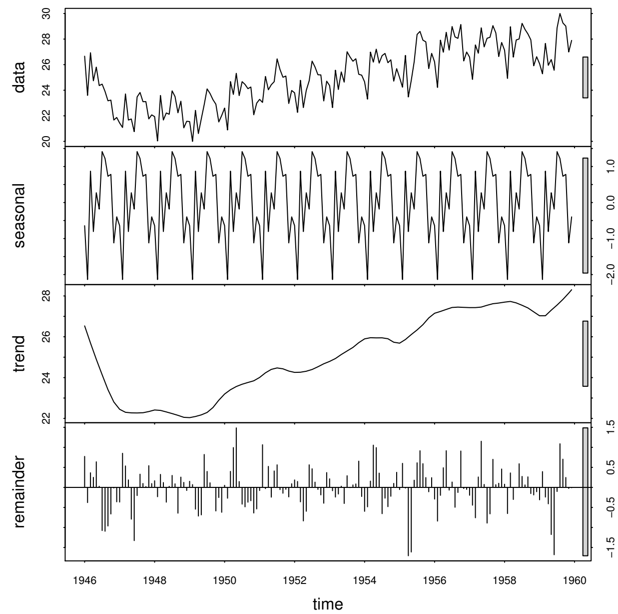
(c) Describe the differences between a Recursive Scheme and a Rolling Window Scheme. How do the interpretation of the results differ from one another?

3.(15%) The plot below shows the number of births per month in New York city, from January 1946 to December 1959.



(a) If you had to choose between an AR(p) and an MA(q) model for this data set, which one would you choose and why?

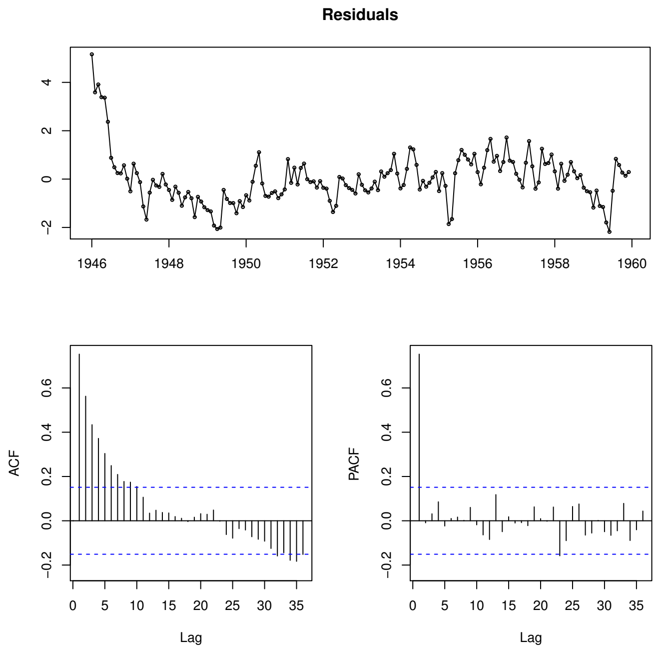
(b) Below we show the `stl` plot of the data. In terms of trend, seasonality, and cycles, briefly discuss which components you would include and why.



(c) Below is the summary from a fit to the data. Discuss what type of model fit was used, its statistical significance, and the behavior of the seasonal factors. Based on your answer, what else would you consider including in your model to improve the current fit?

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	21.4654	0.3237	66.32	0.0000
trend	0.0369	0.0017	21.11	0.0000
season2	-1.5299	0.4140	-3.70	0.0003
season3	1.4426	0.4140	3.48	0.0006
season4	-0.2608	0.4141	-0.63	0.5298
season5	0.7896	0.4141	1.91	0.0584
season6	0.3075	0.4141	0.74	0.4588
season7	1.8733	0.4142	4.52	0.0000
season8	1.6502	0.4142	3.98	0.0001
season9	1.1285	0.4143	2.72	0.0072
season10	1.1573	0.4143	2.79	0.0059
season11	-0.7689	0.4144	-1.86	0.0654
season12	-0.0552	0.4145	-0.13	0.8942

(d) Suppose we try a different fit to the data and obtain the respective `tsdisplay` plot below for the residuals. What model would you propose for the residuals.



4.(15%) Given the time-series below:

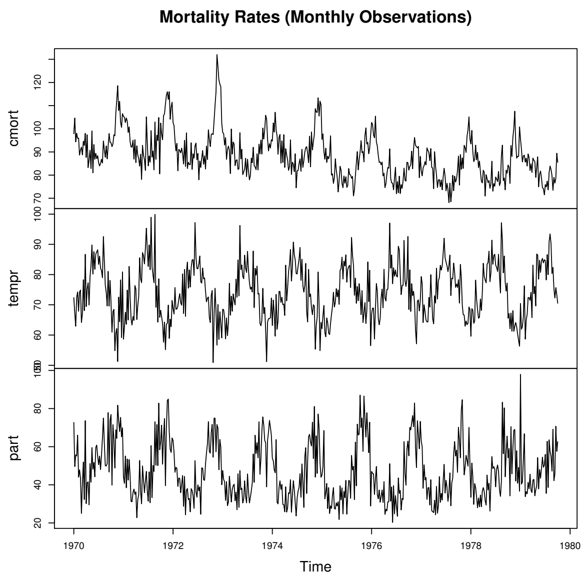
(a) Rewrite the following expression without using the lag operator and identify the process (e.g, AR(p), MA(q), or ARMA(p, q)):

$$\left(L - 2 + \frac{1}{2L - 1}\right) y_t = (L - 1)\varepsilon_t$$

(b) Rewrite the following expression in lag operator form and identify the process (e.g, AR(p), MA(q), or ARMA(p, q)):

$$3\varepsilon_{t-2} - y_{t-2} = 2\varepsilon_{t-1} - 3\varepsilon_{t-3} - 1 + \varepsilon_t - y_t$$

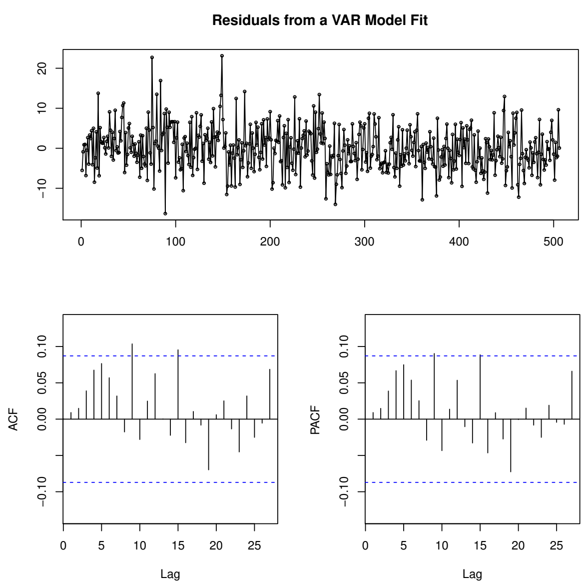
5.(15%) The figure below shows monthly observations of mortality rates (`mort`), pollution (`part`), and temperature (`tempr`). We would like to develop a VAR model to explain mortality rates, and therefore, perform a Granger Causality Test (see table below figure) to determine if temperature and/or pollution Granger cause mortality rates.



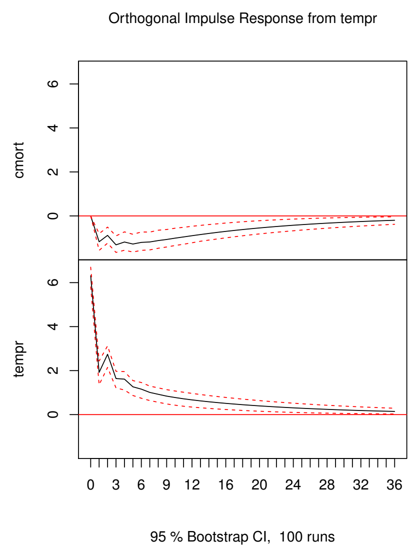

```
> grangertest(cmort˜tempr, 2)
Granger causality test

Model 1: cmort ~ Lags(cmort, 1:1) + Lags(tempr, 1:1)
Model 2: cmort ~ Lags(cmort, 1:1)
  Res.Df Df      F    Pr(>F)
1    504
2    505 -1 75.591 < 2.2e-16 ***
---
Signif. codes:  0 ?***? 0.001 ?**? 0.01 ?*? 0.05 ?.? 0.1 ? ? 1

> grangertest(cmort˜part, 2)
Granger causality test

Model 1: cmort ~ Lags(cmort, 1:1) + Lags(part, 1:1)
Model 2: cmort ~ Lags(cmort, 1:1)
  Res.Df Df      F Pr(>F)
1    504
2    505 -1 2.0487  0.153
```

(a) Based on the Granger test results, what would you conclude in terms of Granger causality for mortality rates?

(b) Below is the `tsdisplay` plot for the residuals from a VAR model fit to Mortality Rates. Interpret the results.



(c) Below is the impulse response function from temperature. Interpret the results.



Questions 6-10 are multiple choice. Please select one answer per question only.

6.(5%) In class we discussed the example: "Which one came first, the chicken or the egg?" Below are the Granger test results. Based on the test which choice is correct?

```
> grangertest(egg ~ chicken, order = 3, data = ChickEgg)
Granger causality test

Model 1: egg ~ Lags(egg, 1:3) + Lags(chicken, 1:3)
Model 2: egg ~ Lags(egg, 1:3)
  Res.Df Df      F Pr(>F)
1     44
2     47 -3 0.5916 0.6238
> grangertest(chicken ~ egg, order = 3, data = ChickEgg)
Granger causality test

Model 1: chicken ~ Lags(chicken, 1:3) + Lags(egg, 1:3)
Model 2: chicken ~ Lags(chicken, 1:3)
  Res.Df Df     F   Pr(>F)
1     44
2     47 -3 5.405 0.002966 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(a) The chicken came first

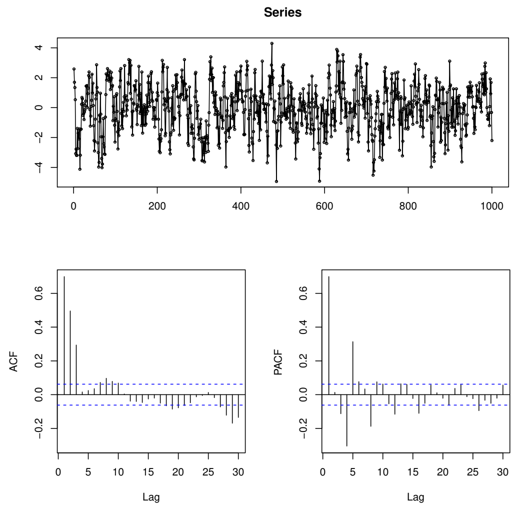
(b) The egg came first

(c) The test suggests that both Granger cause each other

(d) The test suggests that there is no granger causality between the two

(e) None of the above

7.(5%) Which model more closely corresponds to the figure below?



(a) $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + \varepsilon_t$

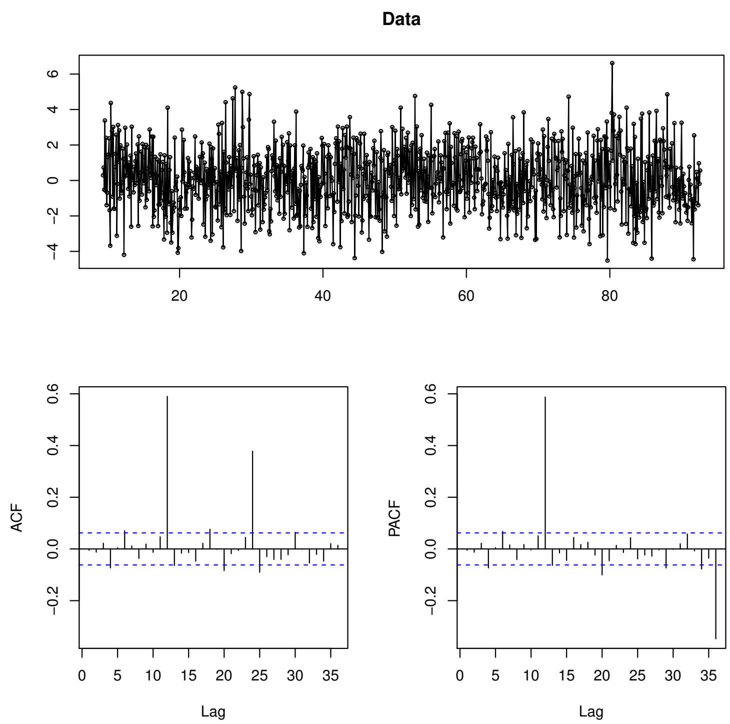
(b) $y_t = \theta_3 \varepsilon_{t-3} + \varepsilon_t$

(c) $y_t = \phi_3 y_{t-3} + \varepsilon_t$

(d) $y_t = \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3} + \varepsilon_t$

(e) None of the above

8.(5%) The data below are monthly observations. Which model more closely corresponds to the figure below?



(a) S-MA(1)

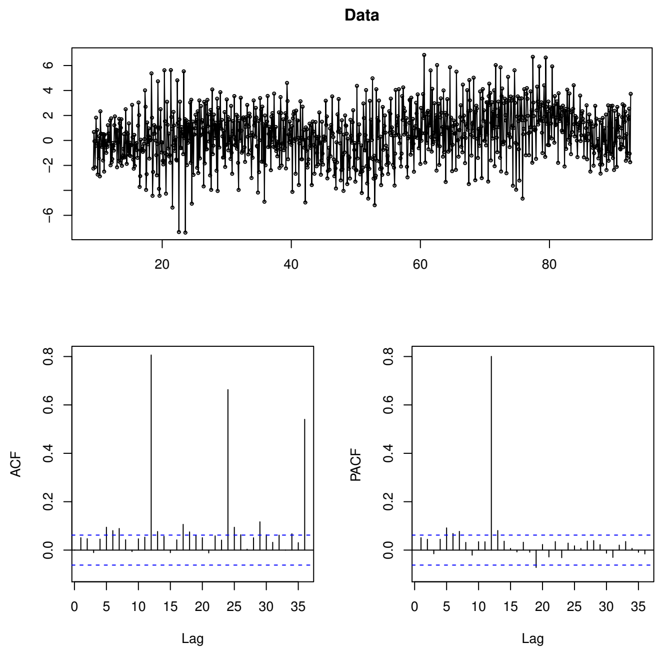
(b) S-AR(1)

(c) S-MA(2)

(d) S-AR(2)

(e) S-ARMA(1,1)

9.(5%) The data below are monthly observations. Which model more closely corresponds to the figure below?



(a) S-AR(1)

(b) S-AR(4)

(c) S-MA(4)

(d) S-MA(1)

(e) None of the above

10.(5%) Assume you are trying to choose a best fit model for a particular dataset. If all the model choices listed below show identical performance in terms of the fit and errors, which one would be the best choice?

(a) AR(7)

(b) ARMA(1,2)

(c) MA(10)

(d) S-ARMA(1,5)

(e) ARMA(1,1) + S-ARMA(2,2)